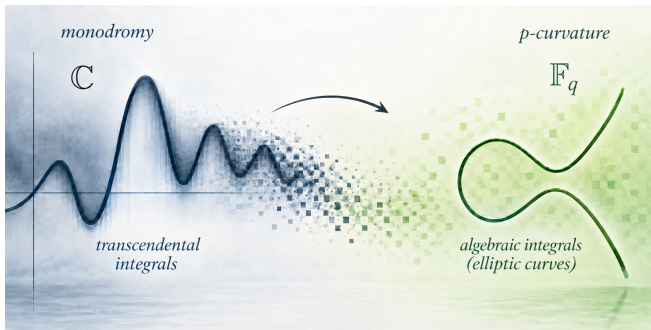


Losing your transcendence: Painlevé equations over finite fields

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Elliptic curves workshop, Oct 2026

Special session at AustMS (1-4 Dec)

Special session at AustMS in Perth: **Cryptography, elliptic curves and dynamics**

Organisers:

Miran Kim (Hanyang University), Tomas Latimer (USYD), Atsuko Miyaji (Osaka University), PR (USYD), Chuanqi Zhang (Monash)

Keynote speakers:

- Craig Costello (QUT)
- Steven Duong (UOW)

Early bird registration closes 16 October

Based on papers:

- (1) *Arithmetic dynamics of a discrete Painlevé equation*,
Nalini Joshi, PR, JPhys A, '26.
- (2) *On integrals of non-autonomous dynamical systems in finite characteristic*
with Nalini Joshi, PR, arXiv:2602.09298
- (3) *On difference-differential Lax pairs and integrals of Painlevé equations in finite characteristic*,
with Nalini Joshi, Tomas Latimer, PR, arXiv:2607.06980.

Plan talk

- 1 Painlevé equations over $\mathbb{C}$
 - Introduction
 - Monodromy $\rightarrow$ transcendental integrals of motion
- 2 Painlevé equations over finite fields
 - Computational experiments
 - p -curvature $\rightarrow$ algebraic integrals of motion
- 3 Future directions

Special functions and linear ODEs

- Many of the classical special functions (Whittaker/Bessel functions, Hermite polynomials etc), satisfy a **linear** second order ODE,

$$y''(z) + a(z)y'(z) + b(z)y(z) = 0.$$

- Linearity** $\implies$ **singularities are fixed**: Singularities of solutions are a subset of singularities of coefficients of ODE.
- For example, solutions of the **Bessel equation**,

$$y''(z) + \frac{1}{z}y'(z) + \frac{z^2 - \alpha^2}{z^2}y(z) = 0,$$

only have singularities at $z = 0, \infty$.

- On the other hand, considering **nonlinear ODEs**,

$$u_{tt} = R(u, u_t, t), \quad R \text{ rational},$$

solutions generally have **movable branch points**, i.e. their locations vary per solution and cannot be read off the ODE itself, e.g.:

$$u_{tt} = -\frac{u_t^2}{u}, \quad \text{solution: } u(t) = c\sqrt{t - \mathbf{d}} \quad (c, \mathbf{d} \in \mathbb{C}).$$

Painlevé equations: hunt for nonlinear special functions

Painlevé, Gambier and Picard (~ 1900), set out to classify all **nonlinear second order ODEs** that share the nice property with linear ODEs that the locations of **singularities** of solutions are **fixed** by the equation.

This eventually led to a complete classification:

$$P_I: \quad u_{tt} = 6u^2 + t,$$

$$P_{II}: \quad u_{tt} = 2u^3 + tu + \alpha,$$

$$P_{III}: \quad u_{tt} = \frac{1}{u}u_t^2 - \frac{1}{t}u_t + \frac{1}{t}(\alpha u^2 + \beta) + \gamma u^3 + \frac{\delta}{u},$$

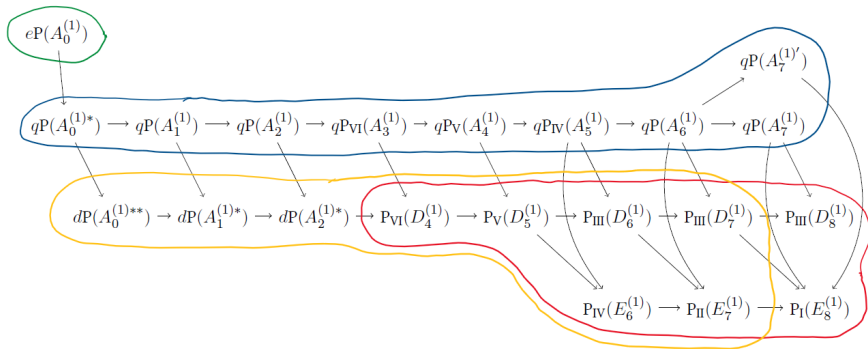
$$P_{IV}: \quad u_{tt} = \frac{1}{2u}u_t^2 + \frac{3}{2}u^3 + 4tu^2 + 2(t^2 - \alpha)u + \frac{\beta}{u},$$

$$P_V: \quad u_{tt} = \left(\frac{1}{2u} + \frac{1}{u-1} \right) u_t^2 - \frac{1}{t}u_t + \frac{(u-1)^2}{t^2} \left(\alpha u + \frac{\beta}{u} \right) + \frac{\gamma}{t}u + \delta \frac{u(u+1)}{u-1},$$

$$P_{VI}: \quad u_{tt} = \left(\frac{1}{u} + \frac{1}{u-1} + \frac{1}{u-t} \right) \frac{u_t^2}{2} - \left(\frac{1}{t} + \frac{1}{t-1} + \frac{1}{u-t} \right) u_t \\ + \frac{u(u-1)(u-t)}{2t^2(t-1)^2} \left(\alpha + \beta \frac{t}{u^2} + \gamma \frac{t-1}{(u-1)^2} + \delta \frac{t(t-1)}{(u-t)^2} \right).$$

Extending hierarchy with discrete Painlevé equations

Sakai (2001) classified all Painlevé equations, **differential** and **discrete**, in terms of their initial value spaces.



- In **green** the elliptic Painlevé equation.
- In **blue** the q -difference Painlevé equations.
- In **yellow** the additive Painlevé equations.
- In **red** the classical Painlevé equations.

No algebraic integrals of motion

Consider the second order differential equation

$$u_{tt} = 6u^2 + a.$$

Multiplying both sides by $2u_t$ and integration yields

$$u_t^2 = 4u^3 + 2au + \mathcal{I},$$

with integration constant $\mathcal{I}$.

We call $\mathcal{I} = u_t^2 - 4u^3 - 2au$ an **integral of motion** of the original ODE.

The first Painlevé equation P_I ,

$$u_{tt} = 6u^2 + t,$$

admits no non-trivial **algebraic** integrals of motion $\mathcal{I} \in \mathbb{C}(u, u_t, t)$.

Much stronger results on **transcendentality** of solutions exist (Malgrange '82, Nishioka '88, '04, Umemura '90).

Transcendental integrals of motion

- The first Painlevé equation governs **deformation** of rank 2 linear ODEs on $\mathbb{CP}^1$ with one irregular singularity of Poincaré rank $5/2$,

$$Y_z = A(z; u, u_t, t)Y, \quad A = z^2 \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 2u \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} -u_t & 2u^2 + t \\ -2u & u_t \end{bmatrix},$$

that **preserves generalised monodromy**.

- Generalised monodromy is encapsulated by 5 **Stokes multipliers**

$$s_k = s_k(u, u_t, t) \quad (k \in \mathbb{Z}_5),$$

each of which is a **transcendental integral of motion** of P_I .

- They lie on an embedded affine del Pezzo surface of degree 5,

$$\mathcal{M} = \{s \in \mathbb{C}^5 : s_k s_{k+1} + s_{k+3} - 1 = 0 \text{ for } k \in \mathbb{Z}_5\}.$$

- The map from solutions of P_I to $\mathcal{M}$ is one-to-one,

$$\text{RH} : \{\text{solutions of } P_I\} \rightarrow \mathcal{M}.$$

- We lose RH when moving to finite fields :(

From complex to arithmetic dynamics

Consider the q -difference first Painlevé equation, $qP_I(A_7^{(1)})$,

$$\begin{cases} \bar{x} = \frac{s t}{s x - y}, \\ \bar{y} = \frac{s x}{y}. \end{cases}$$

It can be considered as a **functional equation**:

- independent variable $t \in \mathbb{CP}^1 \setminus \{0, \infty\}$,
- dependent variables (x, y) ,

$$x = x(t), \quad y = y(t), \quad \bar{x} = x(st), \quad \bar{y} = y(st).$$

- $s \in \mathbb{C}^*$ (usually denoted by q) governs time evolution $t \mapsto s t$.

Alternatively, it can be considered as a (bi-rational) **map**:

$$\phi_{t,s} : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1 \times \mathbb{P}^1, (x, y) \mapsto (\bar{x}, \bar{y}),$$

yielding orbits

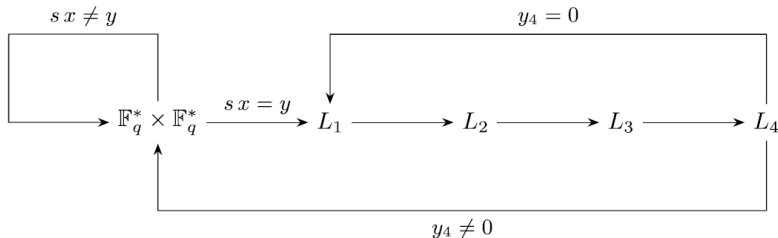
$$\dots \xrightarrow{\phi_{s^{-2}t,s}} (\underline{x}, \underline{y}) \xrightarrow{\phi_{s^{-1}t,s}} (x, y) \xrightarrow{\phi_{t,s}} (\bar{x}, \bar{y}) \xrightarrow{\phi_{st,s}} (\bar{\bar{x}}, \bar{\bar{y}}) \xrightarrow{\phi_{s^2t,s}} \dots$$

Sakai surface

What makes $\phi_{t,s}$ special as a bi-rational map is that it can be **regularised** by a finite (eight) number of blow-ups of $\mathbb{P}^1 \times \mathbb{P}^1$, lifting it to an automorphism of a family of **Sakai surfaces** $\mathcal{X}_{t,s}$.

$$\begin{array}{ccc}
 \mathcal{X}_{t,s} & \xrightarrow{\widetilde{\phi}_{t,s}} & \mathcal{X}_{st,s} \\
 \downarrow \text{Bl}_{t,s} & & \downarrow \text{Bl}_{st,s} \\
 \mathbb{P}^1 \times \mathbb{P}^1 & \xrightarrow{\phi_{t,s}} & \mathbb{P}^1 \times \mathbb{P}^1
 \end{array}$$

This construction works over any field $\mathbf{k}$, in particular a finite one $\mathbf{k} = \mathbb{F}_q$.



*There are some subtleties in characteristic 2 and 3 (Rains,'13)

Example of an orbit

We choose $\mathbf{k} = \mathbb{F}_3$, the evolution parameter $s = 2$ and initial conditions

$$t_0 = 1, \quad x_0 = 1, \quad y_0 = 1.$$

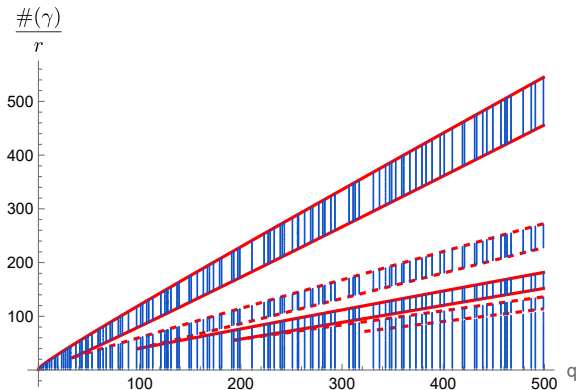
This gives an orbit of length 8.

Time step	t	x	y	Component
0	1	1	1	$\mathbb{F}_3^* \times \mathbb{F}_3^*$
1	2	2	2	$\mathbb{F}_3^* \times \mathbb{F}_3^*$
2	1	2	2	$\mathbb{F}_3^* \times \mathbb{F}_3^*$
3	2	1	2	$\mathbb{F}_3^* \times \mathbb{F}_3^*$
4	1	0	2	L_1
5	2	0	0	L_2
6	1	0	0	L_3
7	2	0	1	L_4
8	1	1	1	$\mathbb{F}_3^* \times \mathbb{F}_3^*$

Computational results

In MAGMA, we computed all orbits ($\sim 200\text{M}$) of qP_I over $\mathbb{F}_q$, with prime power q ranging from 2 to 499. (Joshi, PR '26)

We observed that orbit lengths $\#(\gamma)$ seem **stratified**.



Plot of **orbit lengths** divided by **multiplicative order** $r = o(s)$ of $s \in \mathbb{F}_q^*$.

Matching with Hasse bound

Hasse Bound

Let E be an elliptic curve defined over $\mathbb{F}_q$, then the number of $\mathbb{F}_q$ -rational points on E , $\#E(F_q)$, satisfies

$$|\#E(F_q) - (q + 1)| \leq 2\sqrt{q}.$$

The top band matches perfectly with the **Hasse bound** for elliptic curves,

$$\left| \frac{\#(\gamma)}{r} - (q + 1) \right| \leq 2\sqrt{q}.$$

Subsequent bands take the form

$$\left| \frac{\#(\gamma)}{r} - \frac{(q + 1)}{m} \right| \leq \frac{2\sqrt{q}}{m} \quad (m \geq 2).$$

Are there some elliptic curves hiding in the background?

q-analog of p-curvature

Denote by σ_s operator acting on function $f = f(z)$ by $\sigma_s(f)(z) = f(sz)$.

qP_I governs **deformation** of rank and degree 2 linear q -difference systems with irregular singularities at $z = 0, \infty$ of respective rank 3, 1,

$$\sigma_s(Y) = Y(sz) = A(z)Y, \quad A = A_0 + z A_1 + z^2 A_2,$$

that, over $\mathbb{C}$, preserve “monodromy”.

- Over $\mathbb{F}_q$, s has some finite order $r = o(s)$ and on the field $\mathbb{F}_q(z)$, we have $\sigma_s^r = \text{Id}$.
- However, r -fold application of the q -difference equation yields

$$\sigma_s^r(Y) = M_r(z)Y, \quad M_r(z) := A(s^{r-1}z) \cdot A(s^{r-2}z) \cdot \dots \cdot A(sz) \cdot A(z),$$

and $M_r(z)$ need not be the identity matrix.

- This can be thought of as a **q-analog** of **p-curvature** (Di Vizio, '02).

Over $\mathbb{F}_q$, qP_I governs deformation of the linear system that preserves the **spectrum** of $M_r(z)$.

Coefficients of A matrix

$$A_0 = \begin{bmatrix} t + x - xy & -wx \\ w^{-1}(t + x - ty - 2xy + xy^2) & x(y - 1) \end{bmatrix},$$

$$A_1 = \begin{bmatrix} y - x + x/y - 1 - t/x & w \\ w^{-1}(y - 2x - 1 + xy + x/y - t/x) & 1 \end{bmatrix},$$

$$A_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

Integrals of motion

Joshi and PR, '26

The trace of $M_r(z)$ evaluated at $z = 1$,

$$\mathcal{I}_r(x, y, t, s) := \text{Tr } M_r(1) - (t^r + 1),$$

yields a rational **integral of motion** of qP_I , that is,

$$\mathcal{I}_r(\bar{x}, \bar{y}, s, t, s) = \mathcal{I}_r(x, y, t, s).$$

It is an element of $R[x^{\pm 1}, y^{\pm 1}, t]$ of bidegree $(2r, 2r)$ in (x, y) , where R the ring of r th cyclotomic integers $\mathbb{Z}[s]/(\Phi_r(s))$.

$$\mathcal{I}_1 = -x + y + \frac{x}{y} - \frac{t}{x},$$

$$\mathcal{I}_2 = -\frac{t^2}{x^2} + \frac{2ty}{x} + \frac{2t}{y} - \frac{x^2}{y^2} + \frac{2x^2}{y} - x^2 - 2xy + 2x - y^2,$$

where respectively $s = 1, s = -1$.

Some more integrals

$$\begin{aligned}\mathcal{I}_3 = & -3sx^2y + 3sty - \frac{3sx^2}{y} + 6sx^2 - 3sxy^2 + 3sxy - \frac{t^3}{x^3} + \frac{3t^2y}{x^2} + \frac{3t^2}{xy} - \frac{3ty^2}{x} \\ & - \frac{3tx}{y^2} + \frac{3tx}{y} + \frac{x^3}{y^3} - \frac{3x^3}{y^2} + \frac{3x^3}{y} - x^3 - 3x^2y - \frac{3x^2}{y} + 6x^2 + y^3,\end{aligned}$$

$$\begin{aligned}\mathcal{I}_4 = & -\frac{4stx}{y} + \frac{4st^2y}{x} + 4stx - 8sty^2 + 4sty + \frac{4sx^3}{y^2} - 4sx^3y - \frac{12sx^3}{y} + 12sx^3 \\ & + 4sxy^3 - 4sxy^2 - \frac{t^4}{x^4} + \frac{4t^3y}{x^3} + \frac{4t^3}{x^2y} - \frac{6t^2y^2}{x^2} - \frac{4t^2}{x} - \frac{6t^2}{y^2} + \frac{4t^2}{y} + \frac{4tx^2}{y^3} \\ & - \frac{8tx^2}{y^2} + \frac{4tx^2}{y} + \frac{4ty^3}{x} - 4txy + 4tx - \frac{x^4}{y^4} + \frac{4x^4}{y^3} - \frac{6x^4}{y^2} + \frac{4x^4}{y} - x^4 + 6x^2y^2 \\ & - 12x^2y + 6x^2 - y^4,\end{aligned}$$

where respectively $s^2 + s + 1 = 0$, $s^2 + 1 = 0$.

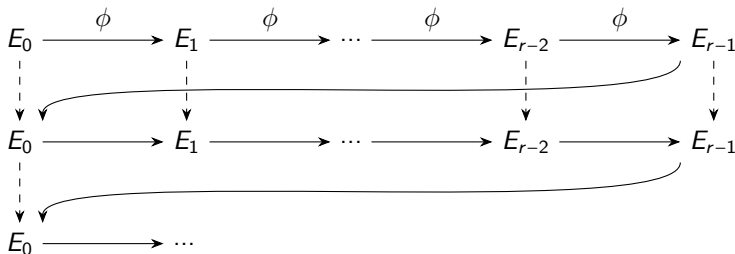
Regarding the example orbit over $\mathbb{F}_3$, with $s = 2$, we have $r = o(s) = 2$ and

$$\mathcal{I}_2 = xy + 2 \left(\frac{t^2}{x^2} + \frac{ty}{x} + \frac{t}{y} + \frac{x^2}{y^2} + \frac{x^2}{y} + x^2 + x + y^2 \right) = 2.$$

Dynamics on fibres

Take $c \in \mathbb{F}_q$ and denote for $m \in \mathbb{Z}/r\mathbb{Z}$ the fibre of $\mathcal{I}$ at time $q^m t$ by

$$E_m = \{(x, y) \in \mathcal{X}_{s^m t, s} : \mathcal{I}(x, y, s^m t, s) = c\}.$$



Joshi and PR, '26

Using MAGMA, we verified that a generic fibre has genus 1 when $1 \leq r \leq 4$. We conjecture it to hold for all $r \geq 1$.

It is very plausible that this can be proven through Sakai theory or extension of it to more general fields (Carstea and Takenawa '12).

p-curvature originates in work of Grothendieck in the 1960ties on understanding differential equations and **monodromy** through arithmetic geometry (local-global principle).

Through p -curvature:

- q -difference linear problems $\rightarrow$ integrals for q -Painlevé equations when s has **finite order**;
- difference linear problems $\rightarrow$ integrals for differential and difference Painlevé equations in **finite characteristic** (Joshi, Latimer, PR, '26);
- differential linear problems $\rightarrow$ integrals for differential and difference Painlevé equations in **finite characteristic**.

type	symbol	operator	p-curvature	condition
differential	$d_z(Y) = Y'(z)$	$D = d_z - A$	$D^p = d_z^p - M_p$	$\text{char}(\mathbf{k}) = p$
difference	$\delta(Y) = Y(z+1)$	$D = \delta^{-1}A$	$D^p = \delta^{-p}M_p$	$\text{char}(\mathbf{k}) = p$
q -difference	$\sigma_s(Y) = Y(sz)$	$D = \sigma_s^{-1}A$	$D^r = \sigma_s^{-r}M_r$	$o(s) = r$

Example: P_I

- Recall **linear problem** for P_I ,

$$Y_z = A(z; u, u_t, t)Y, \quad A = z^2 \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix} + z \begin{bmatrix} 0 & 2u \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} -u_t & 2u^2 + t \\ -2u & u_t \end{bmatrix}.$$

- To compute **p-curvature**, we write $D = d_z - A(z)$ and compute

$$D^p = d_z^p - M_p(z),$$

for some matrix $M_p(z) = M_p(z; u, u_t, t)$, in **characteristic p**.

- Then

$$\mathcal{I}_p(u, u_t, t) := \det(M_p(0))$$

is an **integral of motion** of P_I .

- For example,

$$\mathcal{I}_5 = 2t^5 u^5 + 3t^2 u + tu^3 + tu_t^2 + 4u^{15} + 4uu_t + 4u_t^{10} + 1.$$

- Singular fibres lead to **algebraic solutions** of P_I over $\mathbb{F}_5$,

$$u_{\pm}(t) = \frac{3 \pm 2\sqrt{1+t^5}}{t^2}.$$

Future directions

- ① **Local-global principles** for integrable systems?
- ② New (arithmetic) phenomena over finite fields?
- ③ Can RH be recovered in finite fields?
- ④ Applications in **cryptography**?